MAM Solution Approaches · Independent Review Record

Review method: Each handout is reviewed by an independent sub-agent. The reviewer was not involved in writing, works from a fresh context, and only reads, does not edit. The verification basis includes: the original text of the "Specific behaviours" in each question's marking key in `evidence/`, the full text of the 2016–2025 exam reports, the PDF pages of the original papers and keys, and independent verification using sympy/ scipy.

Handling principles: The editor checks each review comment one by one. All high-severity comments were only modified after returning to the original exam text for verification; anything that could be verified by calculation was checked. For items marked "Doubtful" by the reviewer, changes were made only when supporting evidence could be found.

Conclusion categories: Accepted = modified as per comment; Partially accepted = wording changed but advice not fully followed; Rejected = not changed, with reasons stated.

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01 Differentiation and Its Applications (15 items: accepted 15)

#	Severity	Issue	Handling
1	High	"Not verifying extrema can earn at most half" is fabricated by the editor; the 2024 report does not say this; in 2024A-Q12(c), verification accounts for only 1 of the 4 marks	Accepted: changed to "verification usually carries 1 mark on its own"
2	Medium	"2019 report: one of the worst in the paper, typical error h′(1)" does not match the original report	Accepted: changed to the report's exact wording; h′(1) noted as an editor's reminder; and added 2020 report Q5(c)
3	Medium	"Verification is a guaranteed mark and cannot be omitted" overstates it: the question stems for 2025A-Q12(b) and 2025A-Q14(c) state that verification is not required	Accepted
4	Low	"Omitting one type loses one mark" is not the report's exact wording; question number format is incorrect	Accepted: changed to the actual 4 marking points for 2020F-Q7(d)
5	Medium	"Horizontal point of inflection (f′=f″=0)" may be mistaken for a criterion; counterexample x^4	Accepted
6	Low	"Writing the two-term structure earns marks" is inaccurate; the key requires at least one term correct	Accepted
7	Low	"2023 report: conclusion judged invalid" is not the report's exact wording	Accepted
8	Low	"The report warns not to write it" cannot be found in the MAM reports	Accepted: changed to the actual wording in the 2018, 2019, 2025 reports
9	Low	2025F-Q3 is not a related rates question	Accepted: moved out of Pattern 8
10	Low	Pattern 3 incorrectly cites 2017F-Q6(b)(ii), and omits 2018F-Q3(b), 2019F-Q7(b)	Accepted
11	Low	The reasoning for deriving P″(6)=0 in 2018A-Q15 is not clearly written	Accepted: added "continues increasing afterwards ⇒ horizontal point of inflection", and verified a=−18, b=108
12	Low	"Each condition is worth 1 mark" is an overgeneralisation	Accepted
13	Low	"The derivative does not exist at a turning point" is ambiguous wording	Accepted: changed to "corner (cusp)"
14	Low	The numbering note does not explain 2016S (sample paper)	Accepted
15	Low	"$x^2e^x \to xe^x(x+2)$" and "A″(x)<0" are not clearly expressed	Accepted

03 Motion in a Straight Line (11 items: accepted 11)

#	Severity	Issue	Handling

1	High	"When a = 0, velocity is at an extremum" is an overgeneralisation: $v = t^3$ is a counterexample; in the constant-velocity interval of 2019A-Q12(c), a is always 0; endpoints can also be extrema	Accepted
2	Medium	In "hits 0 and turns back", the meaning of "turns back" is exactly the opposite	Accepted
3	Medium	Piecewise integration is written as "+ −", which is the opposite of the cited 2023A-Q8(c) (that question is first negative then positive)	Accepted: the editor confirmed after verifying $v = -2t \sin t$, changed to "add a negative sign for the $v<0$ interval" and gave the way to write it for that question
4	Medium	Report years are wrong: 2021 was not listed as a major weakness; 2018 and 2023 were omitted	Accepted
5	Medium	The key for 2018A-Q11(c) treats "decelerating 0.5" as $a = -0.5$, which conflicts with the same-sign/opposite-sign rule	Accepted: the editor checked the original question wording and added a reminder note
6	Medium	"Returns to the origin" and "returns to the starting point" are used interchangeably	Accepted
7	Low-Medium	"v”<0 or compare endpoints": questions using calculus must use v” or a sign test	Accepted
8	Low	"Leaves the left side of the screen" is ambiguous	Accepted: changed to "to the right"
9	Low	The graphing requirements for 2025F-Q6(d) are written imprecisely	Accepted
10	Low	"Divide by the inner coefficient" easily leads to miscalculation	Accepted: added "i.e. multiply by 3"
11	Low	Missing citations for 2017A-Q20(c)(d), 2019A-Q12	Accepted

05 Discrete Random Variables (13 items: accepted 13)

#	Severity	Issue	Handling
1	High	2018A-Q12(f) omits the "additional $1 charge"; the new mean should be $1.2\mu + 1$	Accepted: the editor confirmed after checking the original question
2	Medium	"4 decimal places, otherwise marks deducted": the report does not say marks are deducted; the key for 2019A-Q18(c) requires at least 5 places	Accepted: the editor verified by calculation $P \approx 3.0 \times 10^{-5}$
3	Medium	$Y = 3 - X$ is negative when $X = 4$	Accepted: changed to $\max(0, 3 - X)$
4	Medium	"Every year" is untrue; it is actually 6 out of 10 years	Accepted
5	Medium (Doubtful)	Assumptions question: the key only gives "independent" and "same p"; "two outcomes" and "fixed n" are given in the question	Accepted: supported by the key's original text
6	Medium	"binomial and parameters separately worth 1 mark" is ambiguous	Accepted
7	Medium	Pattern 5 incorrectly cites 2022A-Q12 (no binomial sub-question); the single-item probability in 2023A-Q13(b) comes from a uniform distribution	Accepted: deleted the former, replaced it with 2021A-Q8(d), and added a note to the latter
8	Low	$a = \pm 15/22$, with no explanation of why the positive value is taken	Accepted
9	Low	"2020 report Q1" is not labelled F	Accepted
10	Low	Cites 2016S-F4 but does not explain that it is not among the 28 questions	Accepted
11	Low	The marking points for 2017A-Q13 omit the two marks for (d)	Accepted
12	Low	Pattern 6 incorrectly classifies 2021A-Q10(b) and 2023A-Q13(a) as "use binomial"	Accepted

13	Low	"Probability expression separately worth 1 mark" is an overgeneralisation	Accepted: changed to "most questions"

06 Continuous Random Variables and the Normal Distribution (13 items + 1 doubtful: accepted 13, partially accepted 1)

#	Severity	Issue	Handling
1	High	Directly applying $(b-a)^2/12$ for uniform distribution variance will lose marks: the key awards marks for "pdf and domain" and "integral expression", and this formula is not on the formula sheet	Accepted: changed to answering with an integral; formula only for checking
2	Medium	"Keep 4 decimal places" is an overgeneralisation; some questions specify 3 places or exact	Accepted
3	Medium	It reminds students to distinguish cdf, but does not teach the cdf question pattern (2023A-Q11(d)(e))	Accepted: added to Pattern 2
4	Medium	In 2021F-Q6(b)(ii), comparing σ cannot lead to the conclusion; the key uses area	Accepted: changed to the key's reasoning, with a strict bound also given (editor verified 0.383)
5	Low	"Designed according to the 68-95-99.7 rule" is inaccurate	Accepted
6	Low	"2.9 h → 2:54 pm" is not actual exam data	Accepted: changed to 3.8 h → 3:48 pm from 2023A-Q11(c)
7	Low	2025F-Q4 is "Show that k", not "Determine"	Accepted: both questions are cited
8	Low	"Probability expression separately worth 1 mark" is an overgeneralisation	Accepted
9	Low	Unit transformation omits variance, and reverse solving for a and b	Accepted
10	Low	The unit of variance is omitted	Accepted
11	Low	The histogram question pattern omits converting frequencies to probabilities, cumulative tables, and estimating the expected value from class midpoints	Accepted
12	Low	The notation for 2016S-A12 is inconsistent	Accepted
13	Low	Quotation marks were added around an identifying sentence, making a paraphrase look like an original quote	Accepted
Doubtful	—	"2021 report: many did not write uniform" goes beyond the original report text	Partially accepted: changed to the report's actual wording plus the key answer

02 Integration and Its Applications (16 items: accepted 14, partially accepted 2)

#	Severity	Issue	Handling
1	Medium	"FTC is named every year and examined every year" is untrue (not examined in 2019, 2020, 2024; not named in many years' reports)	Accepted: changed to "frequently tested + named multiple times" and listed the years
2	Medium	2021A-Q9 does not involve integration	Accepted: moved out of Pattern 8 and a note added
3	Medium	"The key's 3 marking points are exactly these three steps" is an overgeneralisation	Accepted
4	Medium	Rectangle estimation only covers increasing functions; 2021F-Q7 is decreasing (1/x)	Accepted
5	Medium	"Set the cross-sectional area equal to the volume" is dimensionally incorrect	Accepted: changed to "cross-sectional area × length = volume"

6	Low	2023F-Q5(b) is not in the form ∫f′	Accepted: replaced with 2023F-Q2(a)(iii); Q5(b) listed separately
7	Low	The A(x) rule omits "f changes from negative to positive → local minimum"	Accepted
8	Low-Medium	Omits the question pattern of directly evaluating a definite integral; 2020F-Q3 is not classified	Accepted: added to Pattern 1 (2020F-Q3 is already cited in Pattern 5)
9	Low (Doubtful)	"The report says many lost marks here" is unsupported	Accepted: changed to the actual wording in the 2017 report
10	Low	"Use CAS to calculate exact values" does not match the CF paper	Partially accepted: changed to "find the function then integrate"; trapezium retained
11	Low	The constant c appears from nowhere	Accepted
12	Low	d/dx(x sin 3x) should be 2x sin 3x	Accepted
13	Low	2023F-Q2(c)(iii) is not an area question	Accepted
14	Low	"Split into intervals when crossing the x-axis" is an overgeneralisation	Accepted
15	Low	Missing $n \neq -1$; WACE convention writes ln f(x)	Accepted
16	Low	The explanation of the 2016 sample paper and the question number format	Partially accepted: an explanation was added at the start; the question number format remains `2016S-F1` unchanged

04 Exponentials and Logarithms (12 items: 11 accepted, 1 partially accepted)

#	Severity	Issue	Action
1	Medium	Report year and weak-area correspondence summary is inaccurate	Accepted: list item by item by year
2	Medium	The 2023 report refers to readings (2023F-Q4(c)), not ratios; "cannot be subtracted" conflicts with the key	Accepted
3	Medium	"9 am + 4.7 h ⇒ 1:42 pm" has incorrect values; 2018A-Q9(d) is actually 9.40 h ⇒ 6:24 pm	Accepted: after the editor verified ln1.6/0.05 = 9.40, corrected it, and added rounding down for latest
4	Medium	The ratio formula only wrote /10; 2016S-A13 (k=20), 2016A-Q12 (k=1) and 2023F-Q4 (k=2) differ	Accepted: changed to the general formula L = k·log
5	Medium	"Write decrease rates as positive numbers" is overgeneralised; the key for 2018A-Q9(c) gives a negative number	Accepted
6	Medium-high	Missing the "calculus of e^x / ln x" pattern; 13 questions are unassigned	Partially accepted: added cross-reference Pattern 7; methods are in the differentiation and integration notes to avoid duplication
7	Low	The 2016 sample paper is not explained	Accepted
8	Low	"Almost every question's first step is interconversion" is overgeneralised	Accepted
9	Low	Pattern 1 incorrectly cites 2016F-Q1 and 2025F-Q7(a)	Accepted
10	Low	Missing context rounding rules	Accepted
11	Low	WACE key often writes ln f(x)	Accepted
12	Low	"Dashed line" is not a key requirement	Accepted

07 Sample Proportions and Confidence Intervals (11 items: 11 accepted)

#	Severity	Issue	Action

1	High	Minimum sample size p̂ rule written backwards: the key still uses 0.5 when p̂ is known (2023A-Q7(c), 2024A-Q10(f), 2025A-Q16(e); the 2023 report specifically flags this)	Accepted: after checking the question stems for the three questions, the editor corrected it; under the old rule, 2023A-Q7(c) would give 2263 instead of 2401. The question-setting skill was corrected at the same time
2	Medium	The reminder from the 2023 report was paired with a 2024 example sentence	Accepted
3	Medium	2025A-Q15(f) is a "prove" question; the answer is No, and it was classified incorrectly	Accepted: list "distinguishing the two question types" separately
4	Medium	The example combines time and location into one deviation, violating the rule on the same page	Accepted: split into two
5	Medium	Missing binomial distribution assumption questions (2020A-Q12(h), 2021A-Q11(g))	Accepted
6	Low	"2–4 marks per year / 2 marks per source" is overgeneralised	Accepted
7	Low	2024A-Q10(e) belongs to Pattern 5; Pattern 5 lacks a past-paper list	Accepted
8	Low	The calculation for 9603.6 and 1.96 is inconsistent	Accepted
9	Low	"Switch to a binomial distribution" was incorrectly written as a remedy for (a)	Accepted
10	Low	"Random" is not a key requirement for 2016A-Q10(a)	Accepted
11	Low (doubtful)	Reverse-solving the confidence level omitted 2023A-Q12(c)(d)	Accepted

Summary (7 MAM papers): 92 comments in total (including 1 "doubtful"), 88 accepted, 4 partially accepted, 0 rejected. The 5 high-severity items were all modified after checking against the original paper text.

Classification corrections after user feedback (2026-10-09)

After the user pointed out that unrelated questions were mixed into the vectors question bank, a keyword review was done for all tags, and the key was checked item by item:

- MAM 2021A-Q9 removed the "integration" tag (the whole question only involves solving equations, differentiation, and finding extrema; there is no integration; the integration notes reviewer also pointed this out).
- MAM 2024A-Q10 removed the "discrete random variable" tag (the whole question has no binomial distribution).